Learned Fusion Detection and Selective Purification for Adversarial Images

CSCI 8000 final project proposal | Team 5 | Ruqing Liu, Yifan Zhou, Xinliang Li

Problem description

Our target is a clean-trained CIFAR-10 image classifier based on ResNet-18 [8-10]. Small, deliberately chosen perturbations can change its prediction. Feature squeezing detects suspicious inputs by comparing predictions before and after input transformations; its original multi-squeezer rule takes the maximum disagreement and applies a threshold [1]. Defense-GAN purifies inputs but uses iterative inference [2]. Our goal is to test an input-side defense that learns how to combine squeezing signals and purifies only flagged images. The frozen classifier and CIFAR-10 define the project scope; we make no certified-robustness claim.

Motivation

Our team wants to test whether detecting an attack actually improves the final classification, and whether selective cleaning protects clean images without excessive runtime. These questions connect the course's FGSM, C&W, feature-squeezing, and generative-defense topics to a measurable system-level result. A negative result under unseen or adaptive attacks would also clarify the limits of simple input defenses.

Proposed method and our contribution

Learned detector. Apply 4-bit quantization per color channel, 3 x 3 median smoothing, and JPEG compression (quality 50). For each transformation, compute the L1 difference between the frozen classifier's probability vectors on the original and transformed images; add the original prediction margin. Fit logistic regression on clean and attacked training images. Calibrate its threshold to 5% false positives on clean validation data and compare it with the original maximum-difference rule at the same rate [1].

Selective purifier. Train a small convolutional autoencoder D on adversarial-to-clean pairs. The main run uses $L = L_{rec} + 0.5\, L_{cls} + L_{id}$: mean absolute error between $D(x_{adv})$ and clean x; cross-entropy of the frozen classifier on $D(x_{adv})$ against label y; and mean absolute error between D(x) and x. Each term is batch-averaged and each pixel loss is pixel-averaged. We will ablate the classification term. At inference, the detector routes only flagged images through D.

What is new in this project. Our team will implement the learned fusion score, this three-term purifier training, conditional routing, and a matched-baseline evaluation. These components and their end-to-end analysis extend the prior max-rule and always-purify baselines; we do not claim to have invented feature squeezing or denoising. The detector needs four classifier evaluations per image, so speed is an outcome to measure, not an assumed benefit.

Evaluation and scientific value

Reserve 5,000 of the 50,000 CIFAR-10 training images for validation; keep the official 10,000-image test set untouched until evaluation [8]. Train detector and purifier on FGSM and PGD examples at L-infinity budget 4/255 [3, 4]. Test FGSM/PGD at 4/255 and 8/255; test C&W L2 separately on a fixed, seeded 1,000-image subset and report achieved L2 distortions [5]. Compare no defense, max-rule gating of the same purifier, always-on purification, and learned routing. Report AUROC, detection rate at 5% clean false-positive rate, clean and attacked accuracy, and complete-pipeline latency on identical hardware. We will also check one adaptive PGD surrogate against the full pipeline and verify results on the exact system [6, 7]. Transfer and adaptive results will be labeled separately. The scientific value is a controlled test of whether detection gains translate into useful accuracy after accounting for latency and attack adaptation.

Tasks and timeline

Finish by	Task and deliverable	Proposed lead(s)
Oct 12	CIFAR-10 split, clean ResNet-18, FGSM/PGD attack harness and no-defense baseline	Ruqing
Oct 18	Max-rule baseline and learned fusion detector; three-term purifier prototype	Yifan (detector); Xinliang (purifier)
Oct 22	Integrate routing and produce preliminary FGSM/PGD accuracy and latency for midterm check	All three
Oct 29	C&W and adaptive subset, matched-baseline comparisons, loss/squeezer ablations	Ruqing + Yifan (attacks); Xinliang (ablations)
Nov 3	Final figures, written report, slides, and rehearsal; submit final project Nov 4	All three

Resources and prior work

This is new course work; the team is not reusing prior research results. Public CIFAR-10 [8], the ResNet-18 architecture and TorchVision code [9, 10], and PyTorch/ART attack tools [10, 11] are external resources. One shared GPU is desirable; if unavailable, we will request access or reduce the costly attack subset. All experimental design, integration, and interpretation are the team's work.

References and external resources

[1] Xu, Evans, Qi. Feature Squeezing: Detecting Adversarial Examples in Deep Neural Networks. NDSS 2018.

[2] Samangouei, Kabkab, Chellappa. Defense-GAN: Protecting Classifiers Against Adversarial Attacks Using Generative Models. ICLR 2018.

[3] Goodfellow, Shlens, Szegedy. Explaining and Harnessing Adversarial Examples. ICLR 2015.

[4] Madry et al. Towards Deep Learning Models Resistant to Adversarial Attacks. ICLR 2018.

[5] Carlini, Wagner. Towards Evaluating the Robustness of Neural Networks. IEEE S&P 2017.

[6] Athalye, Carlini, Wagner. Obfuscated Gradients Give a False Sense of Security. ICML 2018.

[7] Tramer et al. On Adaptive Attacks to Adversarial Example Defenses. NeurIPS 2020.

[8] Krizhevsky. Learning Multiple Layers of Features from Tiny Images; CIFAR-10 dataset, 2009. cs.toronto.edu/~kriz/cifar.html.

[9] He et al. Deep Residual Learning for Image Recognition. CVPR 2016.

[10] PyTorch/TorchVision, ResNet-18 implementation and documentation. docs.pytorch.org/vision/stable/models/generated/torchvision.models.resnet18.html.

[11] Nicolae et al. Adversarial Robustness Toolbox v1.0.0. arXiv:1807.01069, 2018.